Bharati Vidyapeeth

(Deemed to be University)

Institute of Management and Entrepreneurship Development, Pune

A REPORT FOR INTERNSHIP PROJECT ON

"E2E-Online-Retail-Sales-Data-Analysis-Project-SQL-Python-PowerBI"

IN THE COMPLETE FULFILLMENT OF THE REQUIREMENT FOR DEGREE OF

MASTER OF COMPUTER APPICATIONS

(SEMESTER IV)

SUBMITTED BY

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Certificate of Completion

This is to certify that **Gaurav Bhagwan Dhamne** (**PRN No: 2403100239**) is a bona-fide student of Semester-IV, Master of Computer Application (MCA) degree program of Bharati Vidyapeeth (Deemed to be University), Pune for the academic year **2024–2026.**

As a part of the University curriculum, he has successfully completed his internship project titled "E2E-Online-Retail-Sales-Data-Analysis-Project-SQL-Python-PowerBI" under the guidance of Dr. Bharati Yelikar.

This project work fulfills the requirement of the university course module "INTERNSHIP PROJECT" of Semester-IV

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DECLARATION

I, undersigned, hereby declare that the project report entitled "E2E-Online-Retail-Sales-Data-Analysis-Project-SQL-Python-PowerBI" is developed and submitted by me in partial fulfillment of the requirement for the award of the degree of Master of Computer Application (MCA), Semester-IV, for the academic year 2024– 2026 under the guidance of **Dr. Bharati Yelikar.**

This project report is based on my own work and has not been submitted elsewhere for the award of any other degree or diploma. All the information included has been collected and analyzed by me during the project work

Student Signature
(Gaurav Dhamne)

Date:

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Lastly, I extend my gratitude to everyone who directly or indirectly contributed to the completion of this project.

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ABSTRACT

This project focuses on analyzing retail sales and return data to understand overall business performance and customer behavior. The data was cleaned and prepared for analysis, followed by exploratory data analysis to identify trends and patterns in sales, returns, and revenue. Key performance indicators such as total sales, return rate, and customer metrics were calculated to evaluate business efficiency. The insights gained from this analysis help in identifying areas for improvement, reducing returns, and supporting data-driven decision-making to enhance profitability

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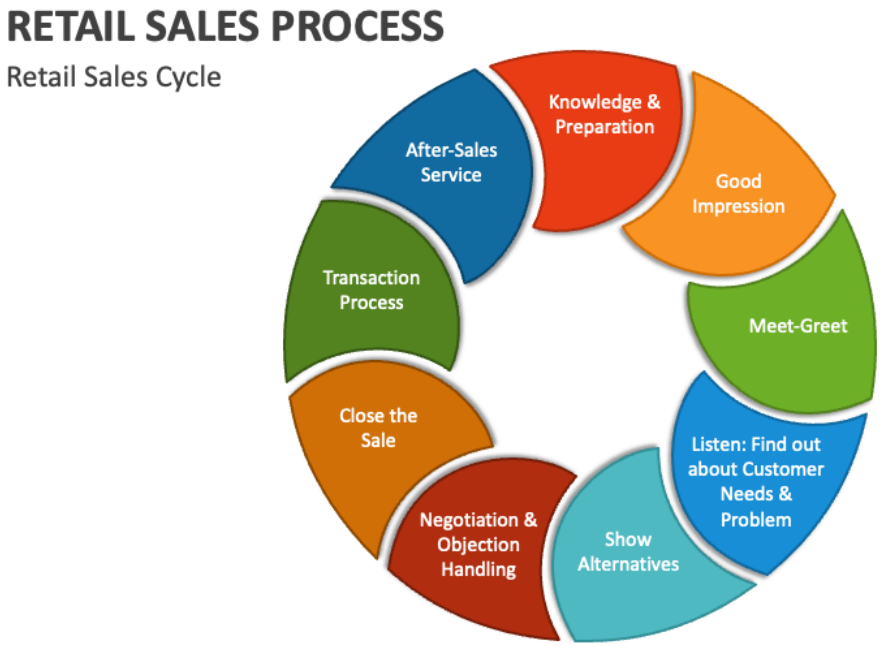
CHAPTER 1 : INTRODUCTION

1.1 INTRODUCTION

The retail industry generates large amounts of transactional data that can provide valuable insights into business performance and customer behavior. This project aims to analyze sales and return data to understand trends, identify key patterns, and evaluate overall performance. By using data analysis techniques, the project focuses on uncovering insights related to sales growth, return rates, and customer activity. These insights help businesses make better decisions, improve operational efficiency, and enhance customer satisfaction.



1.1 Sales Process

1.2 OBJECTIVE

- To analyze sales and return data to understand overall business performance
- To identify trends and patterns in monthly sales and returns
- To evaluate key performance indicators such as revenue, return rate, and customer activity
- To understand customer behavior and identify high-value and high-return customers
- To provide insights that help reduce returns and improve profitability
- To support data-driven decision-making through clear visualizations and dashboards

1.3 MOTIVATION

The motivation behind this project is to understand how data can be used to improve business performance in the retail domain. Sales and return data contain valuable insights that can help identify problems such as high return rates and changing customer behavior. By analyzing this data, the goal is to gain practical experience in data analysis and develop meaningful insights that can support better decision-making, improve customer satisfaction, and increase overall profitability.

1.4 PROJECT OVERVIEW

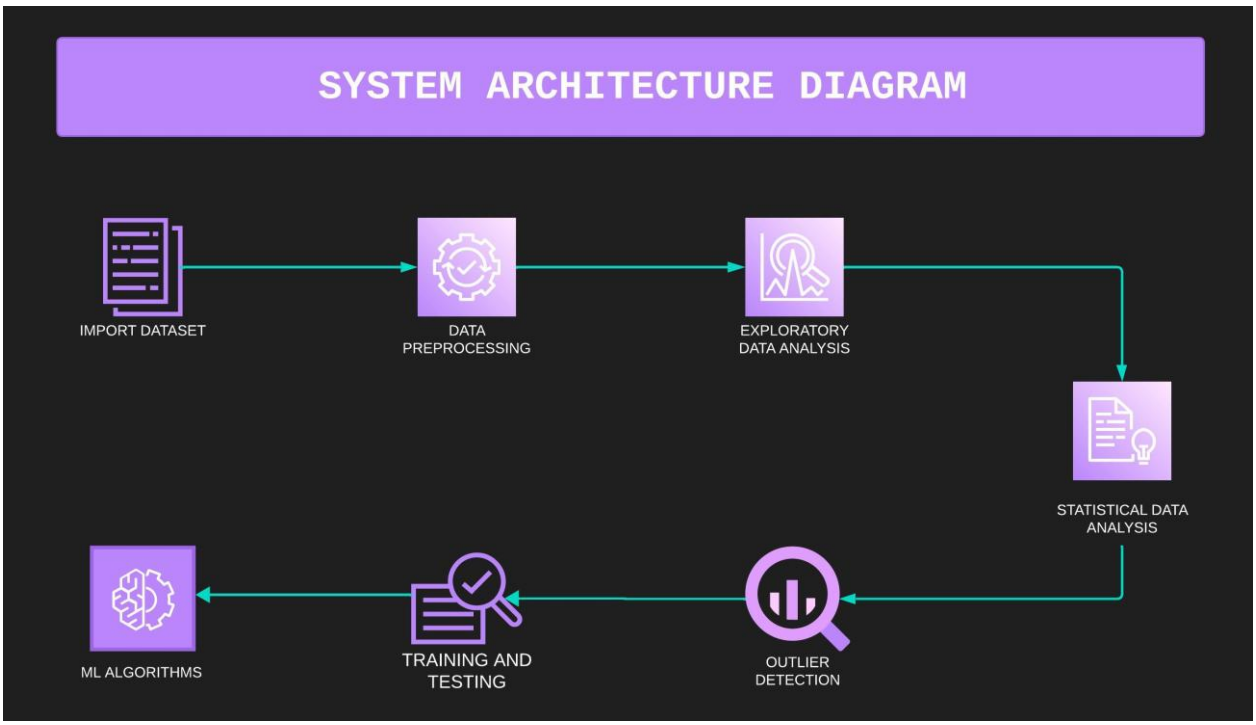
This project delivers a comprehensive, end-to-end analysis of retail sales and returns to evaluate business performance and generate actionable insights. It begins with data ingestion and preparation, where raw transactional data is cleaned, validated, and transformed into a structured format. Key features such as time-based groupings and revenue metrics are engineered to enable deeper analysis.

The exploratory data analysis (EDA) phase uncovers trends in sales, returns, and customer behavior, highlighting seasonal patterns, high-return products, and variations in demand. Building on this, key performance indicators (KPIs) including total sales, return rate, revenue, order volume, and customer metrics are calculated to assess operational efficiency and financial impact.

The findings are presented through clear visualizations and an interactive Power BI dashboard, making the insights easy to interpret and use for decision-making. Overall, the project demonstrates how structured data analysis can help identify inefficiencies, reduce return-related losses, improve customer satisfaction, and support informed business strategies.

CHAPTER 2: SYSTEM ARCHITECTURE

2.1 PROPOSED ARCHITECTURE DIAGRAM



2.1.1 System Architecture

- **Data Source Layer**
  Raw transactional data (CSV Database)
  Contains fields like InvoiceNo, Quantity, UnitPrice, CustomerID

- **Data Ingestion Layer**
  Data is loaded using Python (Pandas)
  Ensures proper extraction and initial validation

- **Data Preparation Layer**
  Cleaning missing values and duplicates
  Creating features like *YearMonth* and *Revenue*
  Splitting data into sales and returns

- **Data Analysis Layer**
  Exploratory Data Analysis (EDA)
  Trend analysis (sales, returns)
  Customer behavior analysis

- **KPI Computation Layer**
  Calculation of metrics like:
  - Sales Quantity
  - Return Rate
  - Revenue
  - Customer metrics
  -
- **Visualization Layer**
  Charts using Matplotlib/Seaborn
  Dashboard creation in Power BI

- **Insight & Decision Layer**
  Business insights generation
  Data-driven decision ma

TECHNOLOGY USED

Technology that would be used in this project are:

- **Python**
  Used for data processing, analysis, and implementing the complete workflow
- **Pandas**
  Used for data cleaning, transformation, and manipulation of datasets
- **NumPy**
  Used for numerical computations and handling arrays
- **Matplotlib & Seaborn**
  Used for data visualization and creating charts for analysis
- **SQL**
  Used for querying, filtering, and retrieving structured data from databases efficiently
- **Jupyter Notebook**
  Used as the development environment for writing code and documenting the project
- **Power BI**
  Used for creating interactive dashboards and visualizing key business insights
- **CSV Dataset**
  Used as the primary data source for analysis

DATASET

The dataset used in this project consists of retail transactional data that includes details of customer purchases and returns. It contains important fields such as InvoiceNo, Quantity, UnitPrice, Country, CustomerID, and InvoiceDate, which help in analyzing sales performance, customer behavior, and return patterns.

The data includes both positive values (sales) and negative values (returns), allowing a clear distinction between successful transactions and returned products. This dataset provides a comprehensive view of business operations and is suitable for performing exploratory data analysis and KPI evaluation.

DATASET OVERVIEW

Attribute	Details
Dataset Name	Online Retail Dataset
Number of Rows	541,909
Number of Columns	8
Data Type	Transactional (Retail Sales Data)

Table 2.1.1: Shows the attribute and his details

COLUMNS DESCRIPTION

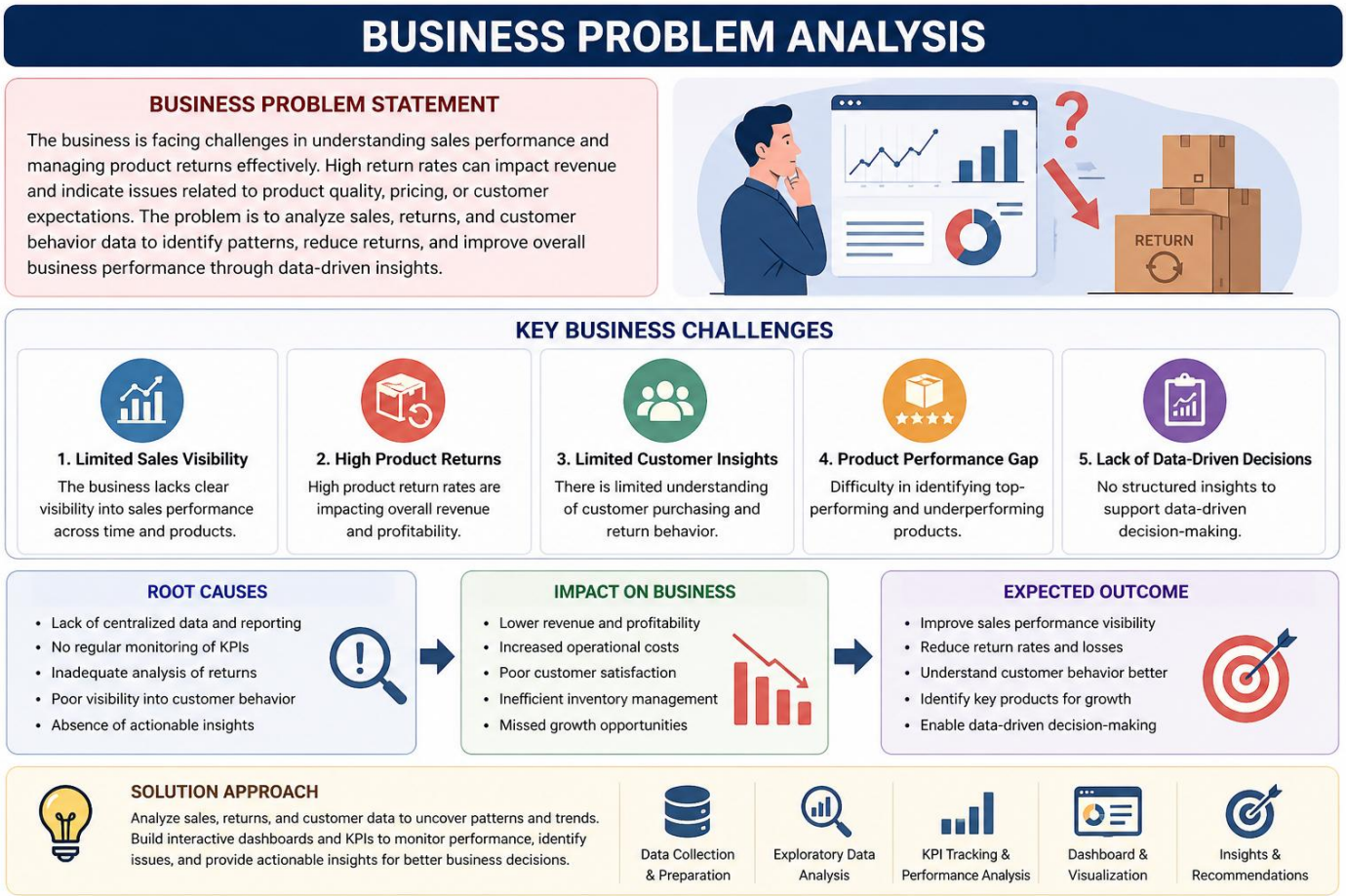
Column Name	Description
InvoiceNo	Unique identifier for each transaction
StockCode	Product/item code
Description	Name of the product
Quantity	Number of items purchased (negative for returns)
InvoiceDate	Date and time of transaction
UnitPrice	Price per unit of product
CustomerID	Unique identifier for each customer
Country	Country where the transaction occurred

Table 2.1.2: Shows the table Column names with description

Inference

The dataset contains detailed retail transaction records with over 500,000 entries, providing a strong foundation for analyzing sales, returns, and customer behavior.

BUSINESS PROBLEM



The business is facing challenges in understanding sales performance and managing product returns effectively. High return rates can impact revenue and indicate issues related to product quality, pricing, or customer expectations. The problem is to analyze sales, returns, and customer behavior data to identify patterns, reduce returns, and improve overall business performance through data-driven insights.

- The business lacks clear visibility into sales performance across time and products
- High product return rates are impacting overall revenue and profitability
- There is limited understanding of customer purchasing and return behavior
- Difficulty in identifying top-performing and underperforming products
- No structured insights to support data-driven decision-making

CHAPTER 3: IMPLEMENTION

3.1 DATA PREPERATION

The data preparation phase involved cleaning and structuring the raw dataset to ensure accuracy and consistency for analysis. Key steps included handling missing values, correcting data types, and creating important features such as YearMonth for time-based analysis and Revenue for financial insights. The data was also split into sales and returns based on quantity, enabling more focused and meaningful analysis of business performance.

Fig. 3.1.1 Data Preperation

```
[5]: df.info()
     df.describe().round(2)

     df.isnull().sum()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 541909 entries, 0 to 541908
Data columns (total 8 columns):
 #   Column       Non-Null Count   Dtype
---  ------       --------------   -----
 0   InvoiceNo    541909 non-null  object
 1   StockCode    541909 non-null  object
 2   Description  540455 non-null  object
 3   Quantity     541909 non-null  int64
 4   InvoiceDate  541909 non-null  object
 5   UnitPrice    541909 non-null  float64
 6   CustomerID   541909 non-null  int64
 7   Country      541909 non-null  object
dtypes: float64(1), int64(2), object(5)
memory usage: 33.1+ MB

[5]: InvoiceNo         0
     StockCode         0
     Description    1454
     Quantity          0
     InvoiceDate       0
     UnitPrice         0
     CustomerID        0
     Country           0
     dtype: int64
```

Fig.3.1.2 Information of Dataset

The dataset contains 541,909 records with mostly complete data, except for some missing values in the **Description** column, indicating minor data cleaning is required

```
[22]: df[df["Quantity"] < 0].sort_values("Quantity").head(10)
```

[22]:

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
460323	540638	72038P	DAMAGES	-990	2011-01-10 12:14:00	0.0	15287	United Kingdom
460434	546023	85175	WRONGLY SOLD SETS	-975	2011-03-08 17:29:00	0.0	15287	United Kingdom
460317	540241	35957	SMALLFOLKART BAUBLE CHRISTMAS DEC	-939	2011-01-05 15:17:00	0.0	15287	United Kingdom
461204	573597	79341	UNSALEABLE, DESTROYED.	-905	2011-10-31 15:18:00	0.0	15287	United Kingdom
461342	580382	16045	CHECK	-900	2011-12-02 17:59:00	0.0	15287	United Kingdom
461255	576368	21135	WET RUSTY	-864	2011-11-14 18:33:00	0.0	15287	United Kingdom
461174	572701	85078	MISSING	-840	2011-10-25 14:31:00	0.0	15287	United Kingdom
461099	569831	20713	WRONGLY CODED-23343	-800	2011-10-06 12:38:00	0.0	15287	United Kingdom
460718	553391	37370	SOLD AS SET/6 BY DOTCOM	-786	2011-05-16 16:42:00	0.0	15287	United Kingdom
461302	578328	72807B	WET PALLET	-696	2011-11-23 17:55:00	0.0	15287	United Kingdom

Fig.3.1.3 Shows dataset interface

- The table shows the **top 10 highest returned or cancelled products**
- Large negative quantities indicate significant returns or damaged/unsellable items
- Many records have descriptions like *DAMAGES*, *WRONG ITEMS*, *MISSING*, highlighting operational or product issues

Final Inference

This project demonstrates a complete data wrangling pipeline...

- Cleaned missing values using group-based techniques
- Removed duplicates and invalid transactions
- Handled outliers using IQR
- Created Revenue columns
- The dataset is now ready for analysis and visualization.

3.2 DATA VISUALIZATION

Fig.3.2.1 Visualization

The visualization of the top returned products highlights items with the highest negative quantities, indicating significant returns or cancellations. A bar chart representation clearly shows which products contribute most to returns, making it easier to identify problem areas.

Products labeled as *damaged*, *wrongly sold*, *missing*, or *unsellable* appear prominently, suggesting issues in product quality, inventory management, or order processing. The visualization helps in quickly spotting patterns and prioritizing corrective actions to reduce returns and improve overall efficiency.

1) Analyze Monthly Sales

Fig.3.2.2 Analyze Monthly Sales

Inference

- The monthly sales trend shows clear fluctuations, with a distinct peak period indicating the highest customer demand and potential seasonality
- Identifying peak and low-performing months helps in better inventory planning, demand forecasting, and targeted marketing strategies

2) Analyze Return Trend Analysis

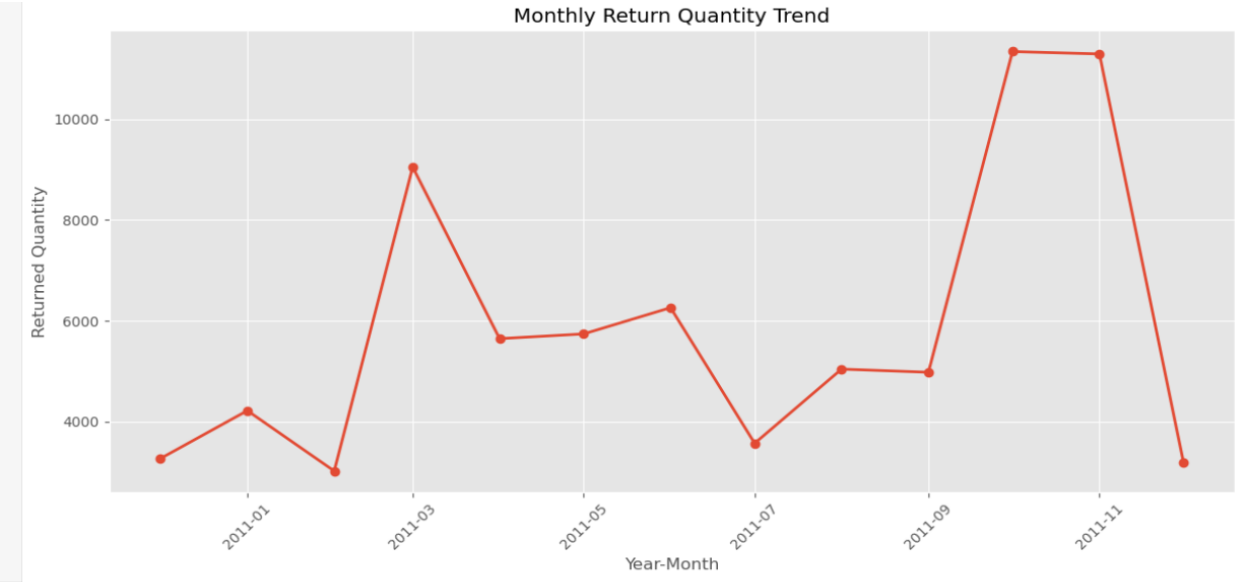


Fig.3.2.3 Analyze Return Trend Analysis

Inference

- The return trend shows noticeable variation across months, with a clear peak indicating periods of higher product returns and potential issues in quality or customer expectations
- Identifying these peak return periods helps businesses take corrective actions to reduce returns and improve overall customer satisfaction

3) Analyze Return Rate in (%)

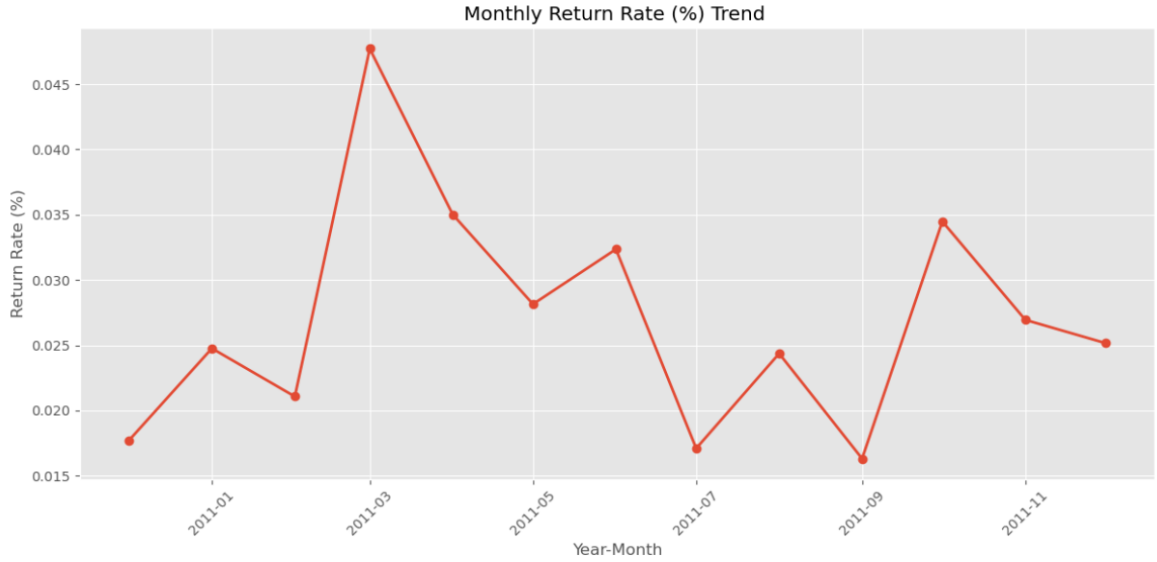


Fig.3.2.4 Analyze Return Rate in (%)

Inference

- The return rate trend highlights periods where returns are disproportionately high compared to sales, indicating potential product or customer experience issues
- Monitoring return rate helps identify abnormal spikes and supports better quality control and business decision-making

4) Analyze Top Selling Products

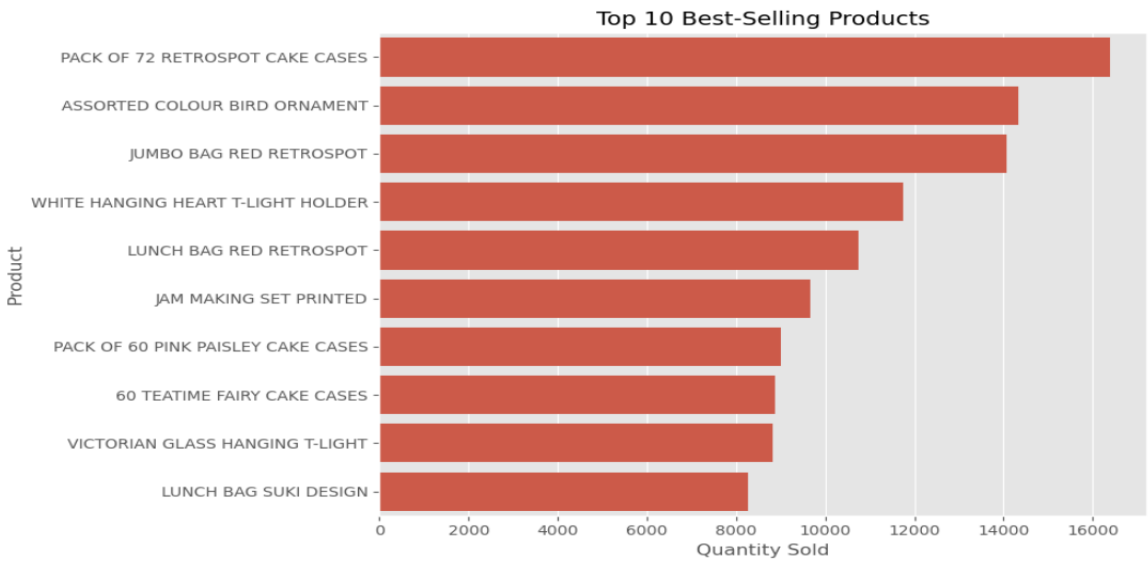


Fig.3.2.5 Analyze Top 10 Selling Prcts

5) Product Return Rate

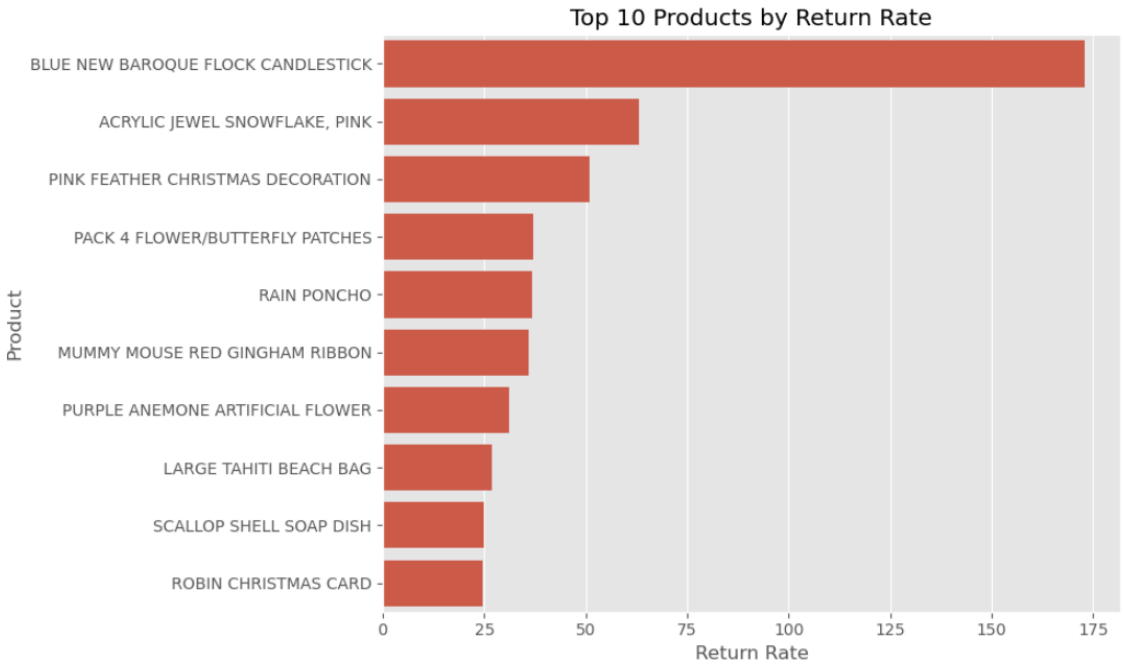


Fig.3.2.6 Analyze top 10 products by return rate

Inference

- Certain products exhibit significantly higher return rates, indicating potential issues in product quality, customer expectations, or descriptions
- Identifying these products helps businesses take targeted actions to reduce returns and improve overall profitability

6) Analyze Monthly Revenue Trends

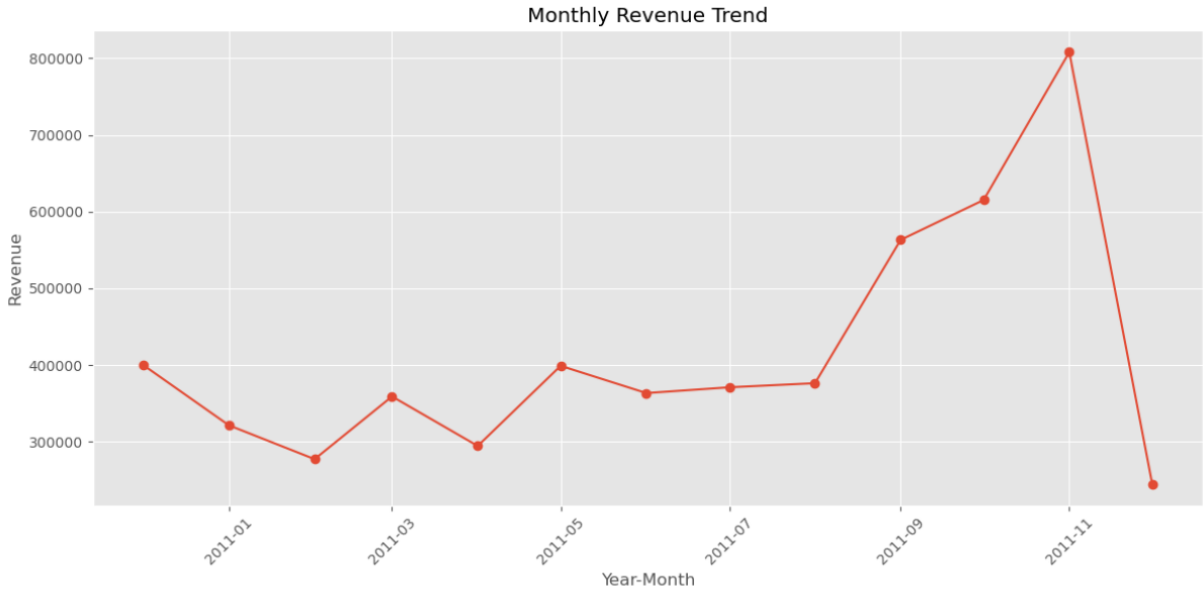


Fig.3.2.7 Analyze Monthly Revenue Trend

Inference

- The monthly revenue trend shows fluctuations over time, with a clear peak indicating the highest earning period and possible seasonal demand
- Understanding these trends helps businesses optimize pricing, marketing strategies, and inventory planning

7) Analyze Sales Vs Return Comparision

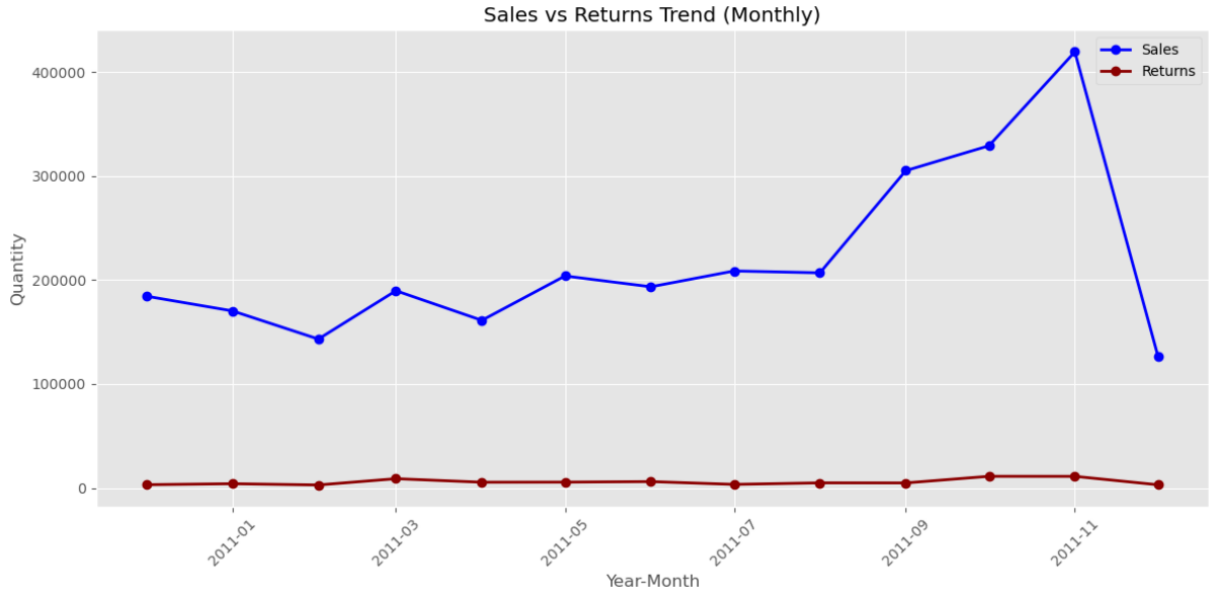


Fig.3.2.8 Analyze Sales Vs Return Comparision

Inference

- The comparison of sales and returns trends highlights the relationship between demand and product returns over time
- Higher return volumes during peak sales periods may indicate issues in product quality or customer satisfaction

8) Analyze Most Returned Products

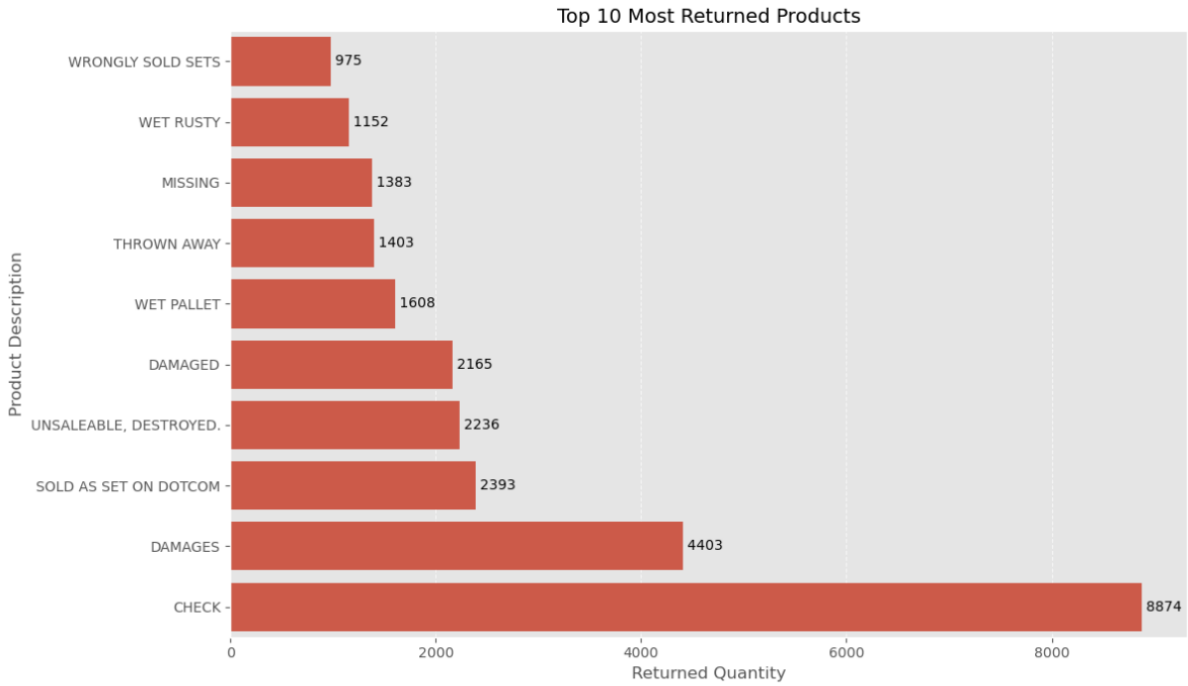


Fig.3.2.9 Top 10 most returned products

Inference

- A small number of products contribute significantly to total returns, indicating potential issues with specific items
- These high-return products should be prioritized for quality checks and customer feedback analysis to reduce return rates

10) Analyze Return Rate Per Product

```
[43]: sales_qty = df[df["Quantity"] > 0].groupby("Description")["Quantity"].sum()
      return_qty = df[df["Quantity"] < 0].groupby("Description")["Quantity"].sum().abs()

      return_rate = (return_qty / sales_qty).sort_values(ascending=False)
      return_rate.head(10)
```

```
[43]: Description
      BLUE NEW BAROQUE FLOCK CANDLESTICK    173.00
      ACRYLIC JEWEL SNOWFLAKE, PINK          63.00
      PINK FEATHER CHRISTMAS DECORATION      51.00
      PACK 4 FLOWER/BUTTERFLY PATCHES        37.00
      RAIN PONCHO                            36.75
      MUMMY MOUSE RED GINGHAM RIBBON         36.00
      PURPLE ANEMONE ARTIFICIAL FLOWER       31.25
      LARGE TAHITI BEACH BAG                 27.00
      SCALLOP SHELL SOAP DISH                25.00
      ROBIN CHRISTMAS CARD                   24.60
      Name: Quantity, dtype: float64
```

Fig.3.2.10 Return rate per product

11) Analyze Customer Analysis

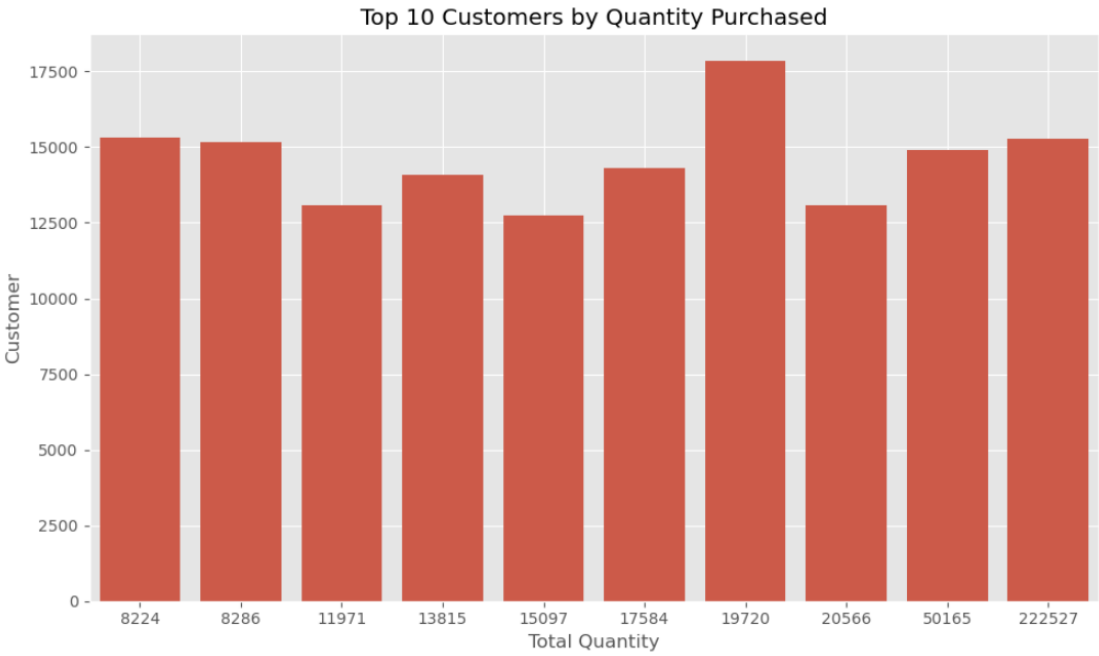
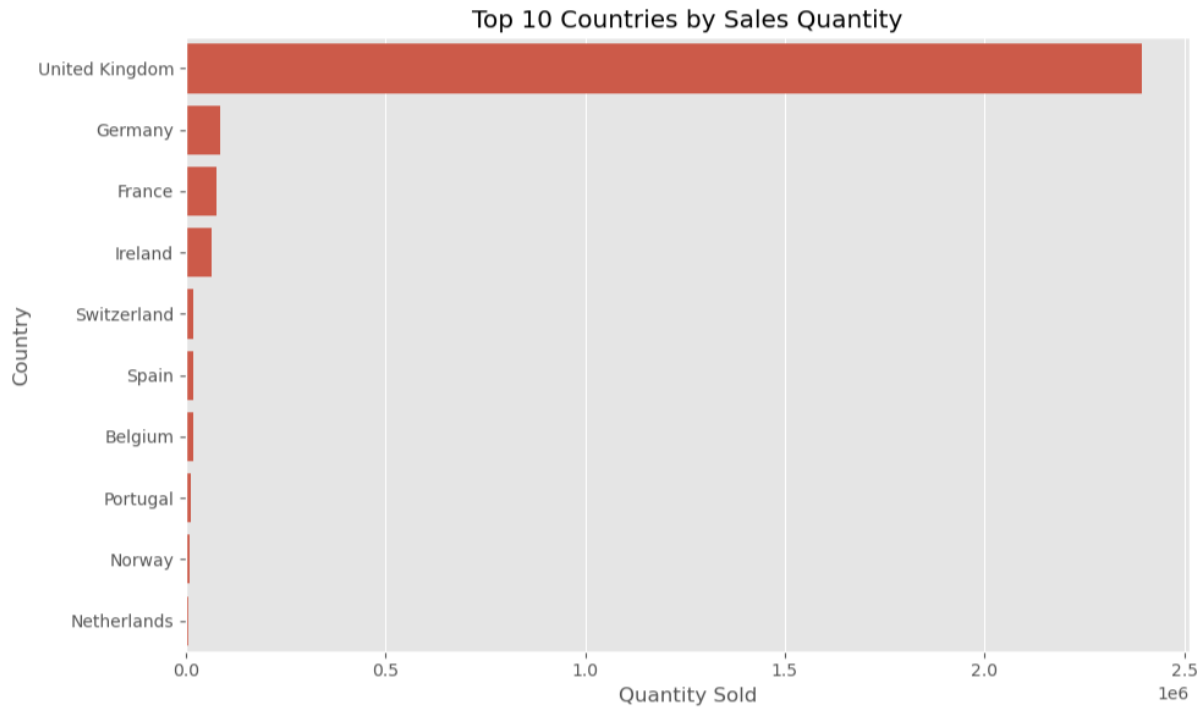


Fig.3.2.11 Top 10 Customer

Inference

- A small group of customers contributes significantly to total sales volume, indicating high-value or bulk buyers
- Identifying these top customers helps businesses focus on retention strategies and personalized engagement

12) Analyze Country Analysis



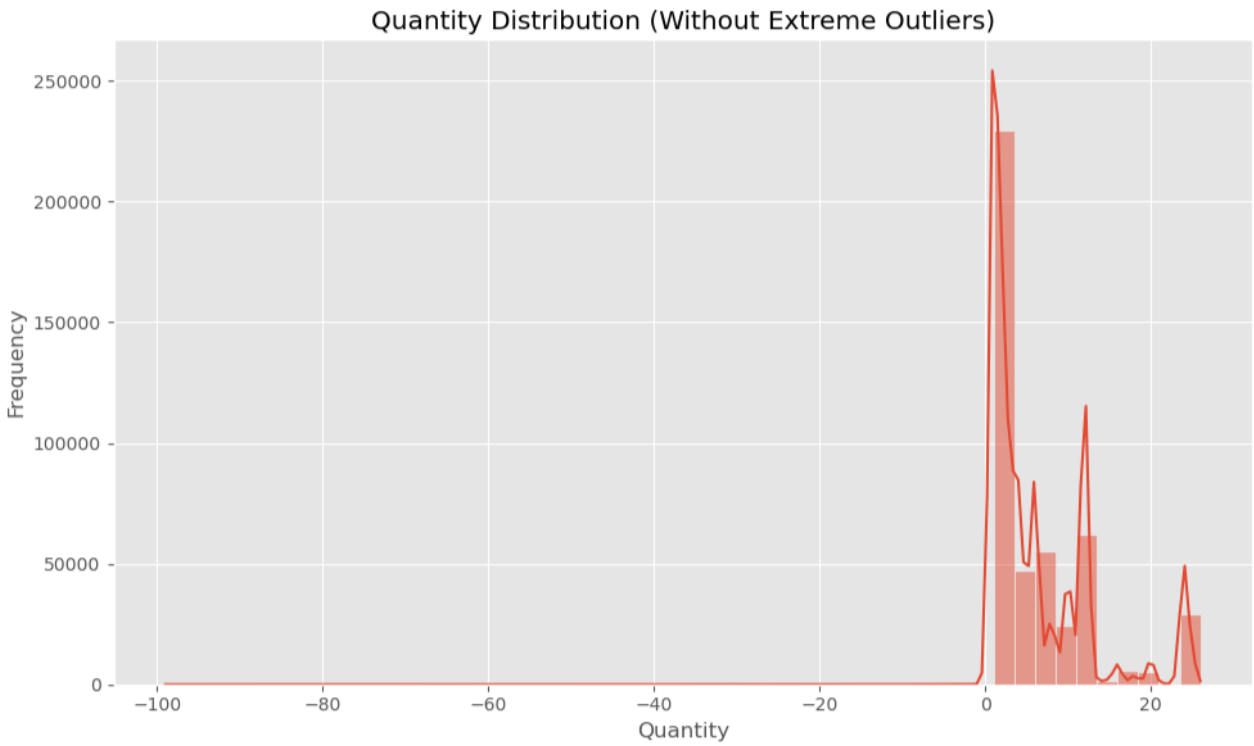

```
[45]: 0    88.493541
      1     3.133570
      2     2.827594
      3     2.418491
      4     0.693300
      5     0.687458
      6     0.659209
      7     0.453363
      8     0.353047
      9     0.280426
      Name: Quantity, dtype: float64
```

3.2.12 Top 10 Countries

Inference

- The analysis shows that a few countries contribute significantly higher sales volume, indicating strong market demand in these regions
- This insight helps businesses focus on high-performing markets while identifying opportunities for expansion in lower-perform

13) Visualize Distribution Analysis

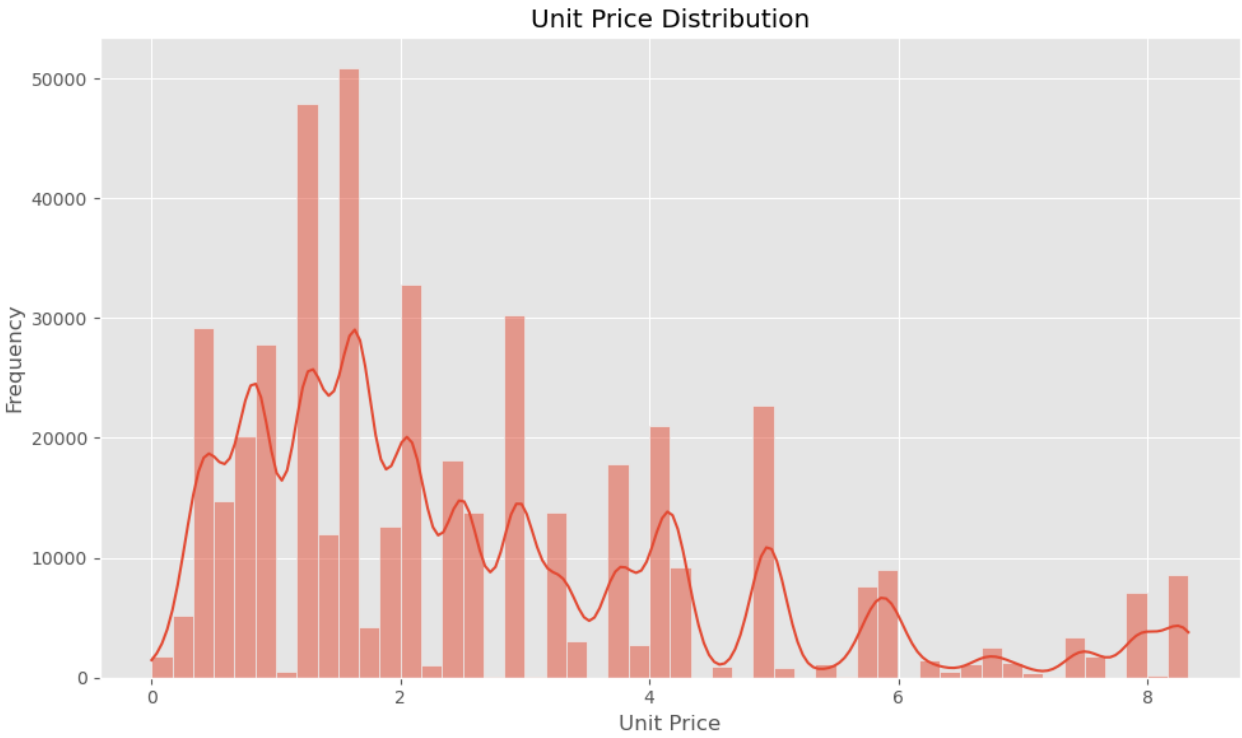


3.2.13 Quantity Distribution

Inference

- The quantity distribution is highly skewed, with most transactions involving small quantities and a few extreme values indicating bulk purchases or large returns
- Negative values represent returns, highlighting the presence of reverse transactions alongside normal sales

14) Visualize Price Distribution



3.2.14 Unit Price Distribution

Inference

- The unit price distribution is right-skewed, indicating that most products are priced at lower ranges with few high-value outliers
- The presence of extreme values suggests premium or bulk-priced products that may impact overall pricing analysis

15) Analyze Unit Price vs Quantity

3.2.15 Unit Price vs Quantity

Inference

- The scatter plot with trend line indicates the relationship between unit price and quantity purchased, showing how demand changes with pricing
- A downward (negative) trend suggests that higher prices tend to reduce purchase quantity, indicating price sensitivity among customers

CHAPTER 4: KPI ANALYSIS AND INSIGHTS

4.1 KPI Analysis

The KPI analysis phase focuses on evaluating key business metrics such as sales, returns, revenue, orders, and customer activity to measure overall performance. By analyzing these indicators, the project identifies trends, efficiency gaps, and the impact of returns on profitability. This helps in making data-driven decisions to improve operations, customer satisfaction, and business growth.

4.1.1 KPI Analysis

The KPI analysis focuses on evaluating key business metrics to understand overall performance and identify areas for improvement. Important indicators such as **total sales quantity, total return quantity, return rate, total orders, total customers, gross revenue, and net revenue** were calculated to measure both operational efficiency and financial impact.

The analysis highlights the relationship between sales and returns, showing how returns affect overall profitability. A higher return rate indicates potential issues related to product quality, pricing, or customer expectations. Revenue metrics help assess business growth, while customer and order metrics provide insights into customer engagement and purchasing behavior.

1) Basic Dataset Overview

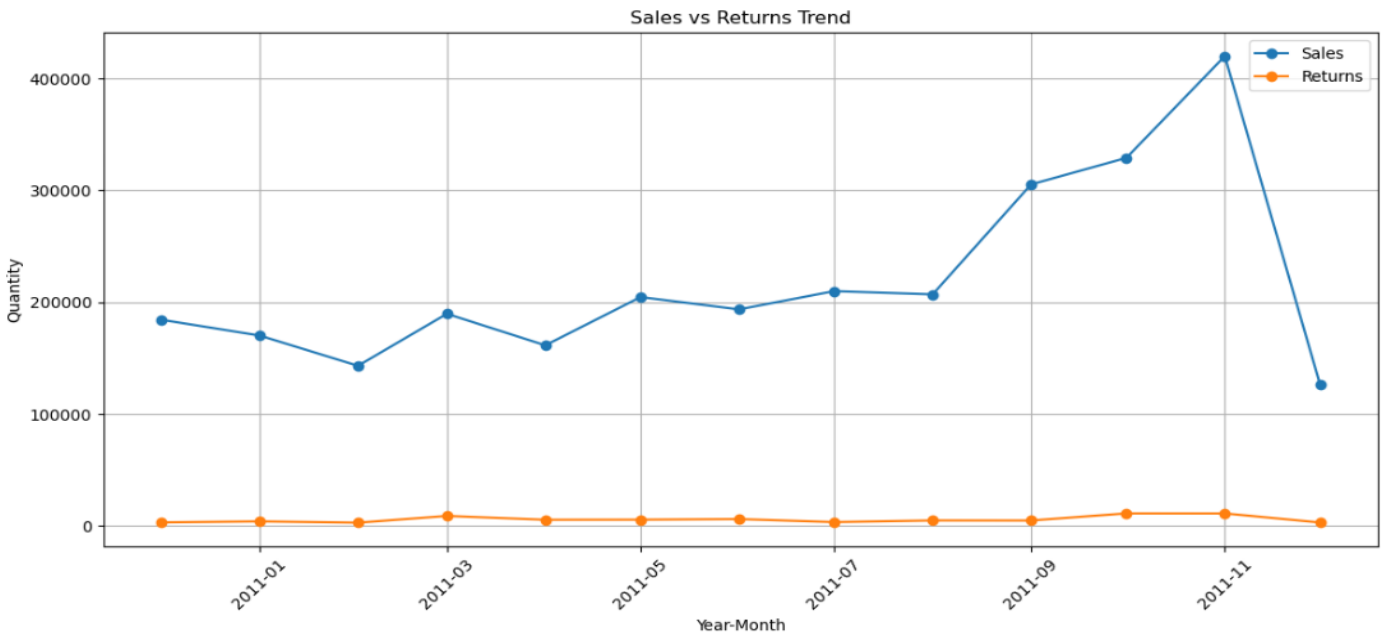
	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	Revenue
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850	United Kingdom	15.30
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850	United Kingdom	20.34
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850	United Kingdom	22.00
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850	United Kingdom	20.34
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	17850	United Kingdom	20.34

4.1.2 Dataset Overview

Inference

- The data shows a single transaction (same InvoiceNo) where multiple products were purchased together by one customer, indicating a multi-item order pattern
- All products have positive quantities and generated revenue, reflecting a successful sale with no returns
- The presence of multiple related items in one order suggests potential product bundling or customer preference for complementary products

2) Monthly KPI



Overall Return Rate

2.69%

4.1.3 Sales vs Return Trend

Inference

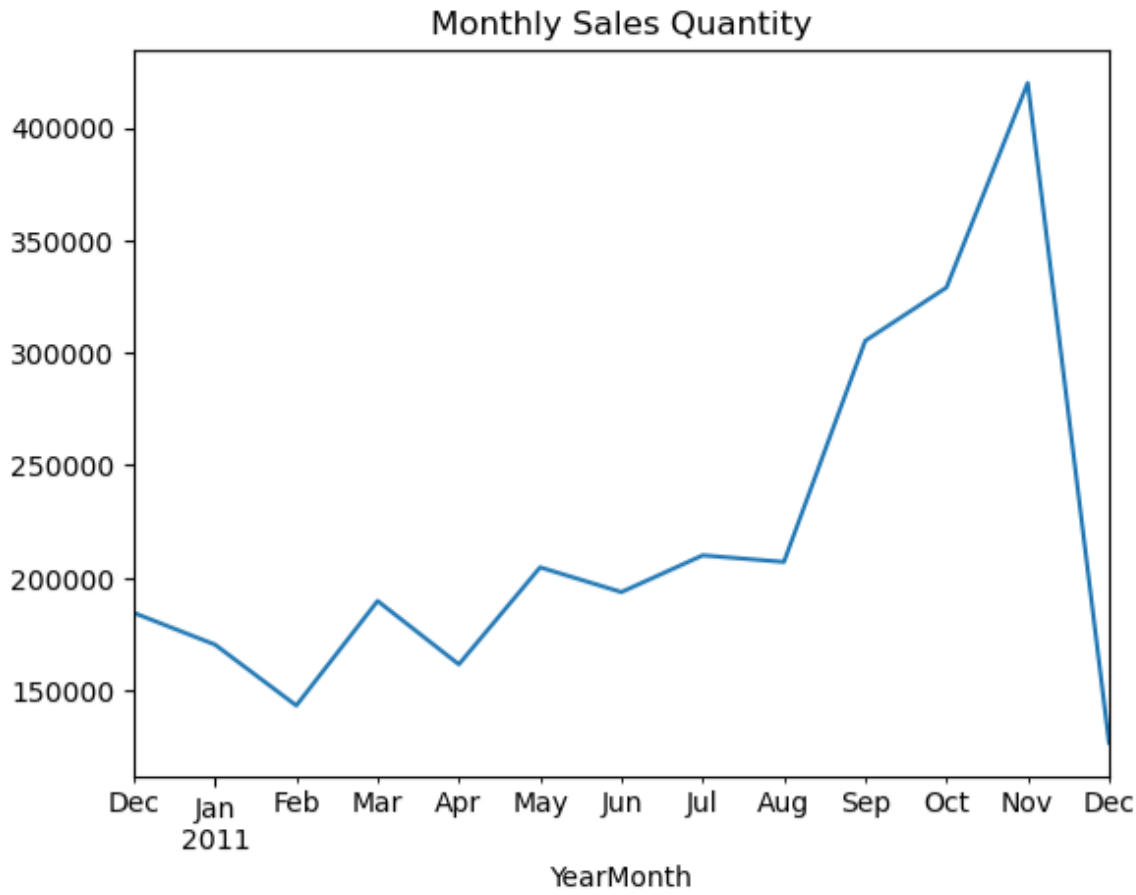
- The monthly return rate reflects the proportion of returned items relative to total sales, serving as a key indicator of product performance and customer satisfaction
- Fluctuations in return rate across months highlight potential seasonal trends or product-related issues that require further investigation

3) Core KPIs

I) Total Sales Quantity = 2845333

xcsx

```
[25]: <Axes: title={'center': 'Monthly Sales Quantity'}, xlabel='YearMonth'>
```



4.1.4 Monthly Sales

Inference

- The total sales quantity represents the overall volume of products sold, indicating the business's transaction scale and demand level
- Higher sales volume reflects strong customer demand and effective product performance

II) Total Return Quantity

Total Return Quantity

76611

```
np.float64(2.6925143735372976)
```

4.1.5 Total Return Orders

Inference

- The total return quantity represents the overall volume of returned products, indicating potential product or service issues
- A high return volume may reflect quality concerns, incorrect product expectations, or operational inefficiencies

III) Total Orders

Total Unique Orders

19393

146.71958954261848

4.1.6 Total Unique Orders

Inference

- The total number of unique orders represents the overall transaction volume, reflecting customer activity and business performance
- A higher number of orders indicates strong customer engagement and consistent demand

IV) Total Customers

Total Unique Customers

4196

```
4.621782650142993
```

4.1.7 Total Orders

Inference

- The total number of unique customers reflects the size of the customer base and overall market reach
- A higher customer count indicates strong acquisition and business growth potential

V) Gross Revenue

Total Gross Revenue

5,398,637.85

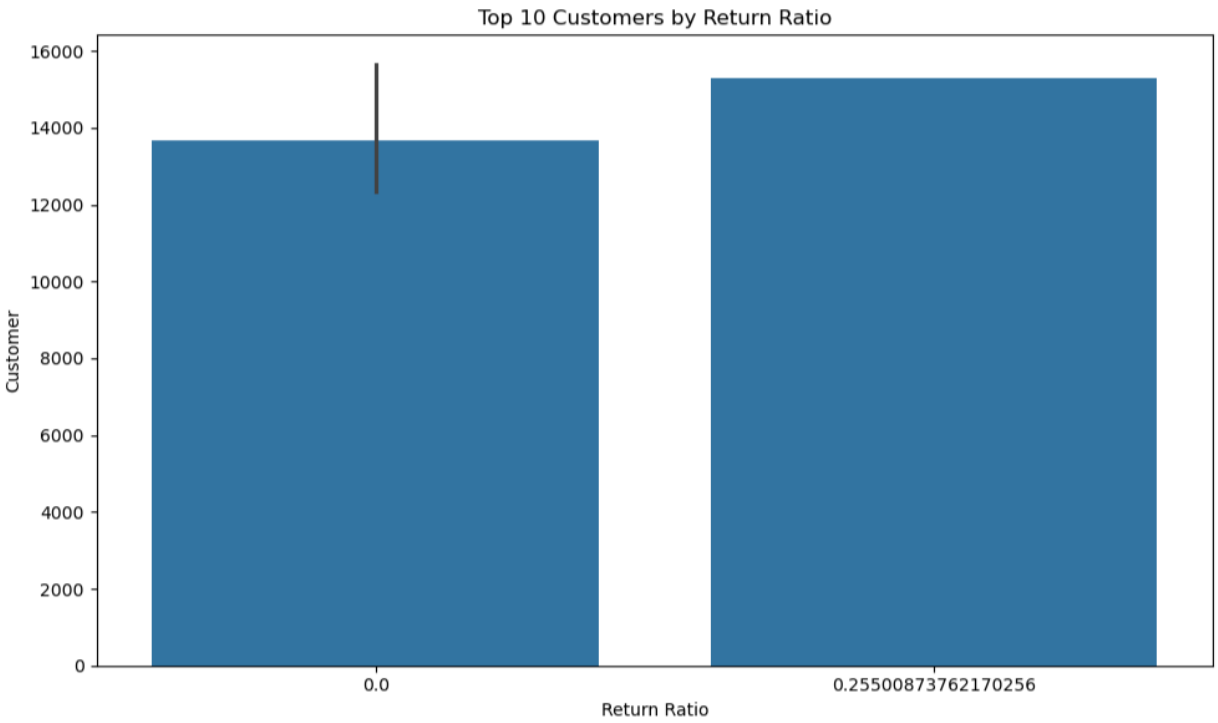
278.38074841437634

4.1.8 Gross Revenue

Inference

- The gross revenue represents the total income generated from sales, reflecting overall business performance and financial health
- Higher revenue indicates strong product demand and effective pricing strategy

4) Customers Return Ratio

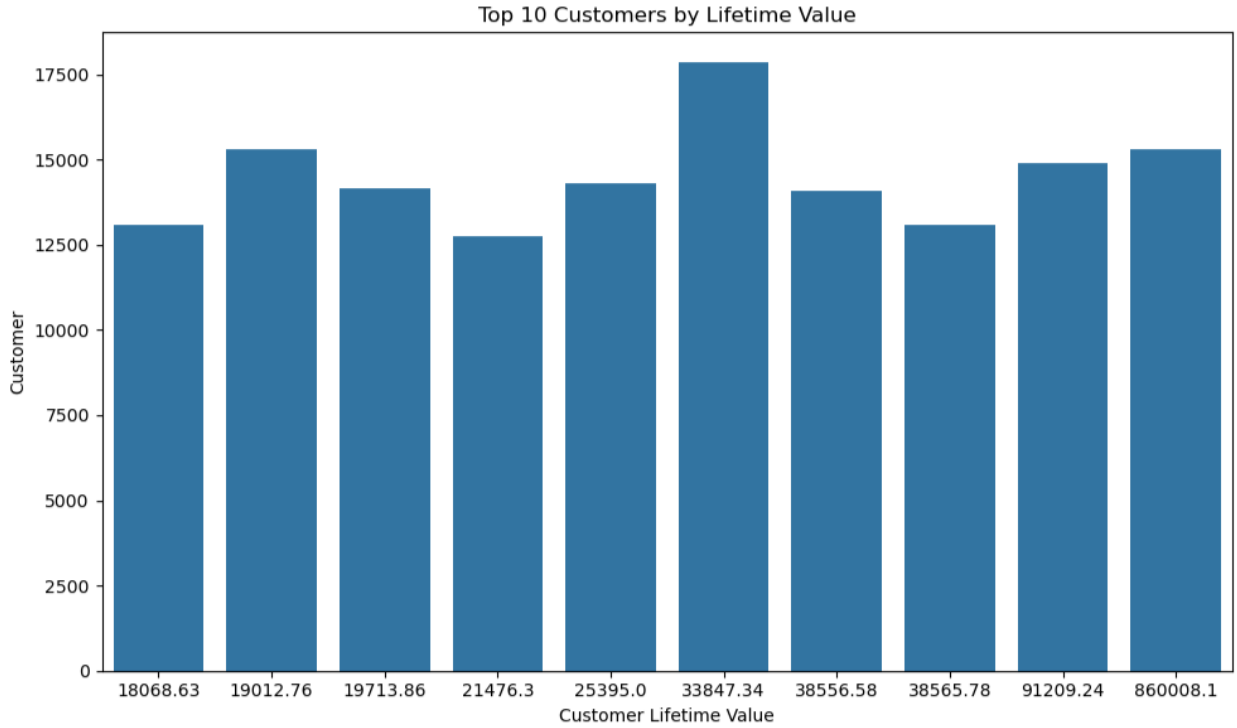


4.1.9 Top 10 Customer

Inference

- The analysis highlights customers with the highest return ratios, indicating patterns of frequent returns and potential dissatisfaction
- Identifying such customers helps in reducing return-related losses and improving customer management strategies

5) Customer Lifetime Value



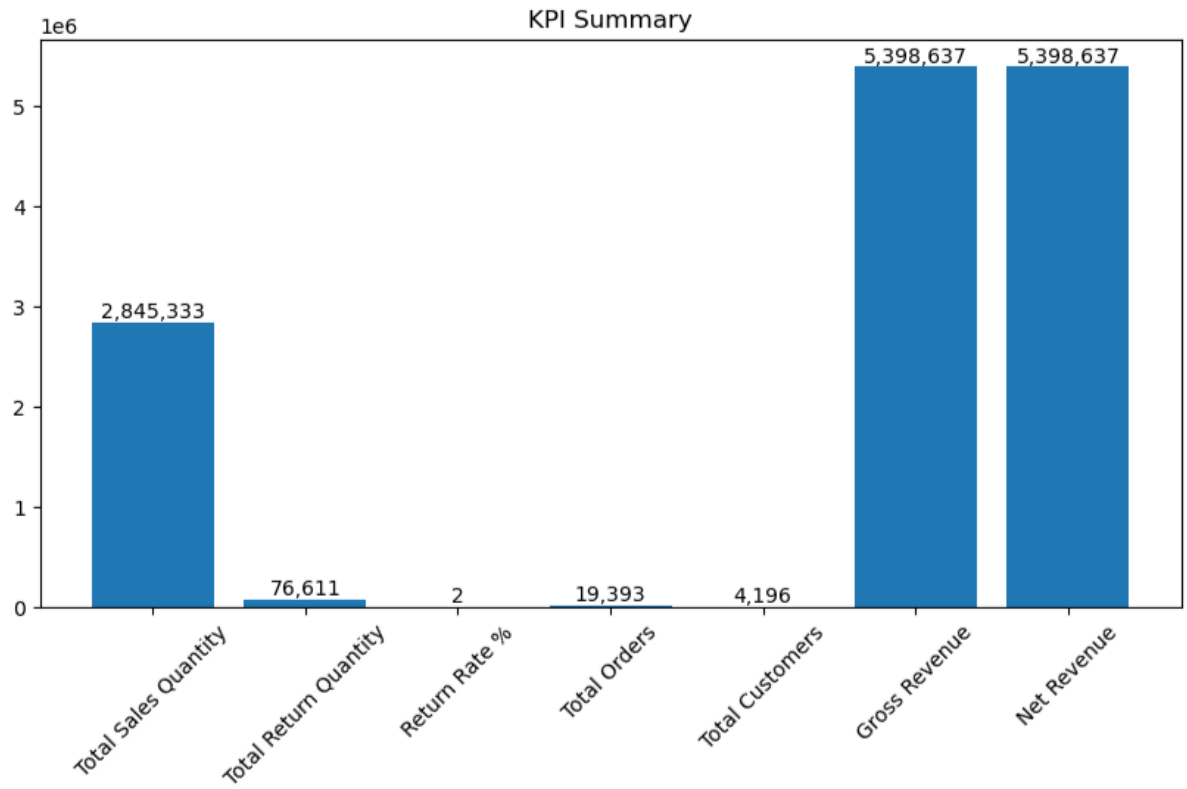

```
[45]: count      4196.000000
      mean       1286.615313
      std       13464.372798
      min           1.900000
      25%         218.070000
      50%         497.810000
      75%        1193.590000
      max      860008.100000
      Name: Revenue, dtype: float64
```

4.1.10 Lifetime Value

Inference

- Customer lifetime value represents the total revenue generated by each customer, helping identify high-value customers contributing most to business revenue
- A small group of customers typically drives a significant portion of total revenue, highlighting opportunities for targeted retention strategies

6) KPI Summary



[69]:

	KPI	Value
0	Total Sales Quantity	2845333.0
1	Total Return Quantity	76611.0
2	Return Rate %	3.0
3	Total Orders	19393.0
4	Total Customers	4196.0
5	Gross Revenue	5398638.0
6	Net Revenue	5398638.0

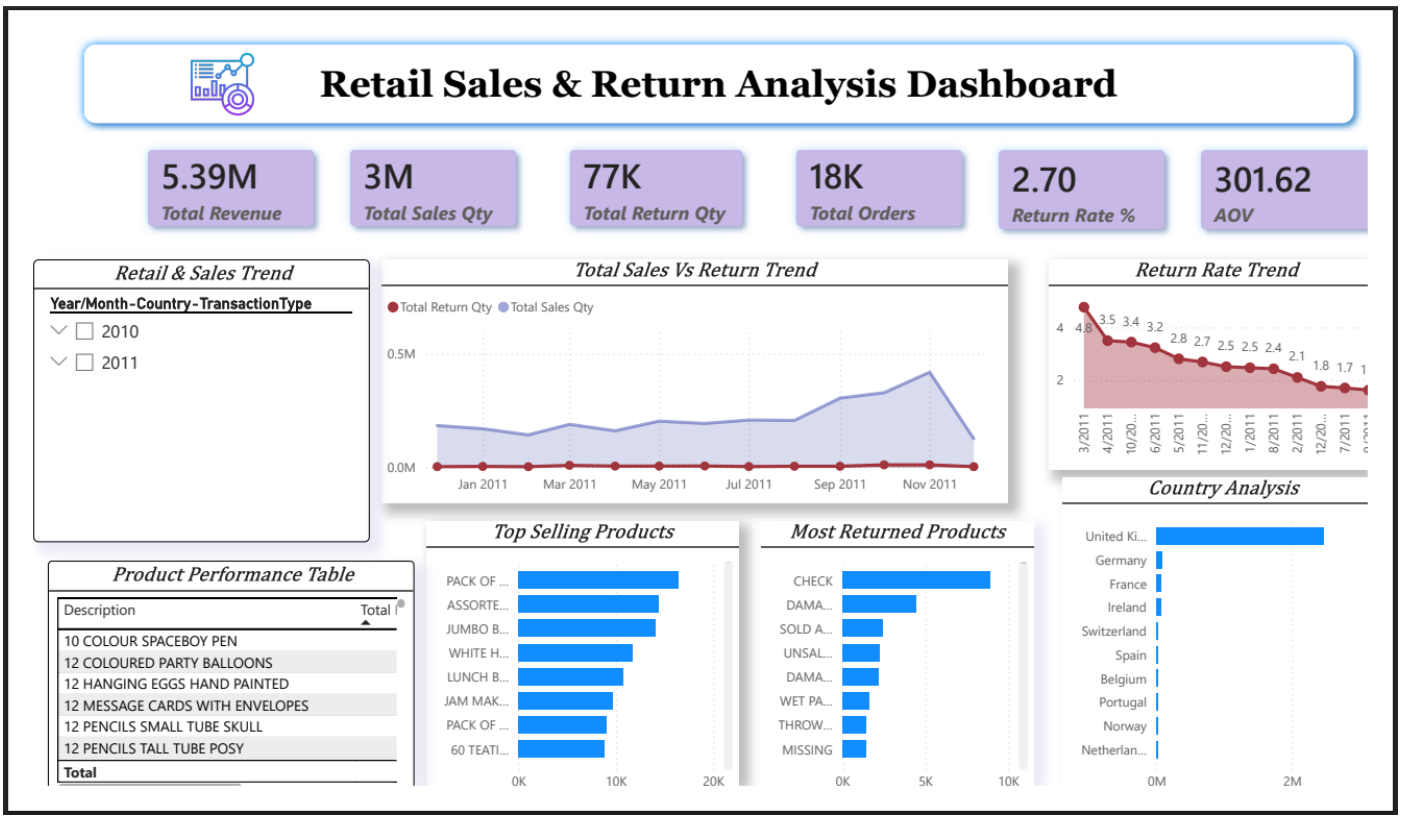
4.1.11 KPI Summary

Inference

- The KPI summary provides a consolidated view of business performance across sales, returns, customers, and revenue
- It highlights key operational metrics and helps identify areas impacting profitability and customer behavior

CHAPTER 5: DASHBOARD VISUALIZATION AND ANALYSIS

5.1 PowerBI Dashboard



View Interactive Dashboard: [Sales_Analysis_Dashboard.pbix]

Dashboard Inference

- The business generated **5.39M total revenue** with **3M sales quantity**, indicating strong overall sales performance
- The **return quantity (77K)** is relatively low compared to total sales, but the **return rate of 2.7%** still impacts profitability
- Sales show a steady trend with peaks in certain months, suggesting **seasonal demand patterns**
- The **return rate is gradually decreasing over time**, indicating improvements in product quality or operations
- A small number of products contribute heavily to returns (e.g., damaged, missing items), highlighting **specific operational issues**
- The **United Kingdom dominates sales**, showing a major concentration of business in one region
- Top-selling products drive significant revenue, while a few problematic products create most of the return-related losses

CHAPTER 6

6.1 CONCLUSION

This project successfully analyzes retail sales and return data to provide a clear understanding of business performance and customer behavior. Through data preparation, exploratory analysis, and KPI evaluation, important patterns in sales trends, return rates, and customer activity were identified.

The analysis shows strong sales performance with relatively controlled return rates, although certain products and operational issues contribute significantly to returns and revenue loss. Insights from the dashboard highlight opportunities to improve product quality, optimize inventory management, and enhance customer satisfaction.

Overall, the project demonstrates the value of data-driven decision-making in identifying inefficiencies, reducing returns, and improving profitability. The use of visualizations and dashboards makes the insights easily understandable and actionable for business growth.

6.2 FUTURE WORK

- Implement **predictive models** to forecast sales and return trends
- Perform **customer segmentation (RFM analysis)** to better understand customer groups
- Develop **real-time dashboards** for continuous monitoring of KPIs
- Apply **machine learning techniques** to identify high-risk return products
- Integrate additional data sources such as customer feedback or reviews for deeper insights
- Improve automation of data pipelines for faster and scalable analysis

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